

A DISTRIBUTION-FREE ANALYSIS OF VARIANCE TECHNIQUE
FOR BLOCK DESIGNS

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Preface

Part I of this paper is due to the first two authors and Part II to all three.

1. Introduction and Summary

Major contributions to the study of distribution-free analysis of variance techniques for complete or incomplete block designs for a one-way classification of treatments have been made by Friedman (1937), Kendall (1948), Durbin (1951), Benard and Van Elteren (1953) and Van Elteren and Noether (1959). Reinach (1965) presented a distribution-free method of analysis for a multi-way classification of treatments in a randomised block design. In all these techniques, which are based on ranks only, the observations in each block are ranked (in increasing order) and ranks allocated independently of the rankings in the other blocks. This same principle of independent rankings in each block will be used in the present paper. Further important contributions have been made by Hodges and Lehmann (1962), Lehmann (1964), Mehra (1964) and Greenberg (1966).

The a.r.e. (asymptotic relative efficiency) of Friedman's test for the RBD (randomised block design) or of Durbin's test for the BIBD (balanced incomplete block design) relative to the appropriate F-test for normal alternatives of shift is equal to $3t/\pi(t+1)$ where t denotes the block size (cf. Van Elteren and Noether (1959)), and thus tends to $3/\pi$ as $t \rightarrow \infty$. Mehra (1964) constructed a test with an a.r.e. value of $3/\pi$ in the above case for the BIBD by using Wilcoxon's test statistic for symmetry applied to all pairs of treatments.

In Part I of the present paper Benard and Van Elteren's procedure for a one-way classification of treatments in a general block design is extended to the more general case where the ranks are replaced by an almost arbitrary function of the ranks. A test statistic is defined which is shown to be distributed asymptotically, when the number of blocks tends to infinity, as a non-central χ^2 -distribution under a class of a.r.e.-type alternatives. It is further shown that the test based on this test statistic, when using a special function of the ranks, has in comparison with the appropriate F-test a greater a.r.e. value than Friedman's (for the RBD) or Durbin's (for the BIBD) test against specific alternatives of shift (for block size > 2) and is obviously easier to apply than Mehra's procedure when the block size becomes large.

Whereas the RBD (with $c(> 1)$ observations per cell) is considered as a special case in Part I, the BIBD is considered as a special case in Part II.

More specifically, in Part II a two-way classification of treatments is considered in a BIBD. The test statistics, which again are based on an almost arbitrary function of the ranks allocated to the

observations in each block, for the two components into which the treatments are subdivided and for the interaction are shown to have, under a class of a.r.e.-type alternatives, asymptotically independent non-central χ^2 -distributions as the number of replications of each treatment combination tends to infinity. Finally, the single degree of freedom approach and the a.r.e. values of the tests with respect to the appropriate parametric F-tests are considered.

P A R T I

2. The general test statistic

In the problem of m rankings (cf. Kendall (1948)) each of m observers ranked k given items. Denote by r_{sj}^* the rank allocated by the s -th observer to the j -th item. The set of ranks $\{r_{sj}^*\}$ can be arranged in a scheme consisting of m rows and k columns with one entry per cell.

If, in the above scheme, the items denote measurements due to different treatments and the observers denote different blocks containing k plots each to which the different treatments have been applied (in a random way), we have in fact a RBD. In such a scheme there is only one observation or measurement in each cell and Friedman's (1937) test is applicable. In practice, however, it often occurs that the number of observations in each cell is not necessarily equal to one but can be any positive integer or zero. This more general case is considered by Benard and Van Elteren (1953). Suppose the (s,j) -th cell contains $k_{sj}(>0)$ observations, i.e. the j -th treatment is applied to k_{sj} plots in the s -th block, and denote the result of the h -th replication of the j -th treatment in the s -th

block by x_{sjh} . The $k_s = \sum_{j=1}^k k_{sj}$ observations in the s -th block are

ranked independently of the observations in the other blocks and ranks allocated, denoting by r_{sjh} the rank allocated to x_{sjh} , so that $1 \leq r_{sjh} \leq k_s$. Then Benard and Van Elteren (1953) defined

$$\tilde{u}_{sj} = \begin{cases} \sum_h [r_{sjh} - \frac{1}{2}(k_s + 1)] & \text{if } k_{sj} > 0 \\ 0 & \text{if } k_{sj} = 0, \end{cases}$$

and they considered a test statistic based on a function of the treatment totals $\tilde{u}_j = \sum_s \tilde{u}_{sj}$, which is shown to have asymptotically, for $m \rightarrow \infty$, a χ^2 -distribution under the null hypothesis of no treatment effects (which implies the hypothesis that every permutation of the ranks in each block is equally likely).

We now consider for every k_s the function $\psi_s(\delta) \equiv \psi_{k_s}(\delta)$ which is assumed continuous and monotonically strictly increasing in δ on the interval $0 < \delta < 1$. Then, for every positive $\epsilon < \frac{1}{2}$, there exists a finite real number L^{**} such that $|\psi_s(\delta)| < L$ for $0 < \epsilon \leq \delta \leq 1 - \epsilon < 1$. Let further $\delta_{sjh} = r_{sjh} / (k_s + 1)$ for all s, j and h .

If $F_{sj}(x)$ denotes the c.d.f. of the population from which the k_{sj} observations of the j -th treatment in the s -th block are drawn,

the following class of hypotheses is considered:

(1) $H : F_{sj}(x) = F_s(x + d_j \cdot m^{-1/2})$ for all real x where the d_j 's are arbitrary finite real numbers and all F_s are assumed continuous.

As null hypothesis is considered:

$H_0 : F_{sj}(x) = F_s(x)$ for all real x and all s and j , i.e. $d_j = 0$ for all j .

It is assumed that no ties occur among the observations in each block, which hold with probability one under the class of hypotheses H considered in this paper. All results which hold true under H_0 , however, remain valid under the more general null hypothesis that all permutations of the observations within each block are equally likely. (In this formulation ties among the observations are admissible although it is then necessary to make certain obvious adjustments - cf. Lemmer (1964)).

Define further

$$(2) \quad t_{sj} = \begin{cases} \sum_h \psi_s(\delta_{sjh}) & \text{if } k_{sj} > 0 \\ 0 & \text{if } k_{sj} = 0, \end{cases}$$

$$(3) \quad \tilde{t}_{sj} = t_{sj} - E(t_{sj} | H_0)$$

and

$$(4) \quad t_j = m^{-1/2} \sum_s \tilde{t}_{sj}.$$

Denote the matrix of the variances and covariances under H_0 of $(k-1)$ of the t_j 's, say $t_1, \dots, t_{k-1}$, by Λ which for the moment is assumed to be non-singular, i.e. $|\Lambda| \neq 0$.

As test statistic we define (cf. Lemmer (1964))

$$(5) \quad T = \sum_{\xi=1}^{k-1} \sum_{\xi'=1}^{k-1} \frac{|\Lambda_{\xi\xi'}|}{|\Lambda|} t_{\xi} t_{\xi'},$$

with $|\Lambda_{\xi\xi'}|$ the cofactor of $\text{cov}(t_{\xi}, t_{\xi'} | H_0)$ in $|\Lambda|$. T can also be written: $T = \vec{t}' \Lambda^{-1} \vec{t}$, where $\vec{t}' = (t_k, \dots, t_{k-1})$ and Λ^{-1} denotes the inverse matrix of Λ .

Remark:

All rows with $k_{sj}=0$ for any $k-1$ cells (s, j) , $j=1, 2, \dots, k$, are deleted since for such a row $\tilde{t}_{sj}=0$ for all j .

3. Second order moments of the t_j under H_0

Let

$$\psi_{sjh} = \psi_s(\delta_{sjh}) \text{ and } \bar{\psi}_s = k_s^{-1} \sum_j \sum_h \psi_{sjh}, \text{ then}$$

^m Throughout part I of this paper it is assumed, except when otherwise stated, that the indices $s, s'=1, 2, \dots, m$; $j, j'=1, 2, \dots, k$; $k, k'=1, 2, \dots, k_{sj}$ and $\xi, \xi'=1, 2, \dots, k-1$.

^{xx}) The symbol L will be used as a generic finite positive constant.

Lemma 1

$E(t_j|H)$ is given by equation (16) if (14) holds.

Proof: Let

$$P_p(x) \stackrel{\text{def}}{=} P(\underline{r}_{sjh} = p | \underline{x}_{sjh} = x, H), \quad *$$

then

$$\begin{aligned} E(\psi_{sjh}|H) &= E(\psi_s(\frac{r_{sjh}}{k_s+1})|H) \\ &= \sum_{p=1}^{k_s} \psi_s(\frac{p}{k_s+1}) P(\underline{r}_{sjh} = p|H) \\ &= \sum_{p=1}^{k_s} \psi_s(\frac{p}{k_s+1}) \int_{-\infty}^{\infty} P_p(x) dF_{sj}(x). \end{aligned}$$

Similar as in Stoker (1955) p. 95-96 by means of Van Dantzig's method of collective marks (cf. Van Dantzig (1947-1950)), it easily follows for $0 < A < 1$ that

$$\sum_{p=1}^{k_s} P_p(x) A^{p-1} = [AF_{sj}(x) + 1 - F_{sj}(x)]^{k_{sj}-1} \prod_{\substack{j'=1 \\ (\neq j)}}^k [AF_{sj'}(x) + 1 - F_{sj'}(x)]^{k_{sj'}},$$

From (1) and (14), however, it follows that

$$\begin{aligned} F_{sj'}(x) &= F_s(x + d_j, /m^{\frac{1}{2}}) = F_{sj}\{x + (d_j, -d_j)/m^{\frac{1}{2}}\} \\ &= F_{sj}(x) + m^{-\frac{1}{2}}(d_j, -d_j)f_{sj}(x) + o(m^{-\frac{1}{2}}) \end{aligned}$$

so that

$$\begin{aligned} \sum_{p=1}^{k_s} P_p(x) A^{p-1} &= [AF_{sj}(x) + 1 - F_{sj}(x)]^{k_{sj}-1} \cdot \\ &\cdot \prod_{\substack{j'=1 \\ (\neq j)}}^k [AF_{sj'}(x) + 1 - F_{sj'}(x) + m^{-\frac{1}{2}}(d_j, -d_j)\{AF_{sj'}(x) - F_{sj'}(x)\} + o(m^{-\frac{1}{2}})]^{k_{sj'}}, \\ &= [AF_{sj}(x) + 1 - F_{sj}(x)]^{k_s-1} + \\ &+ m^{-\frac{1}{2}} \sum_{\substack{j'=1 \\ (\neq j)}}^k k_{sj'}(d_j, -d_j)\{AF_{sj'}(x) - F_{sj'}(x)\}\{AF_{sj}(x) + 1 - F_{sj}(x)\}^{k_s-2} + o(m^{-\frac{1}{2}}) \end{aligned}$$

*) Random variables will, when necessary, be distinguished from other variables or numbers by underlining the random variables.

Since we have $P_p(x) = \text{coefficient of } A^{p-1}$, we obtain

$$\begin{aligned} E(\Psi_{sjh}|H) &= \sum_{p=1}^{k_s} \Psi_s\left(\frac{p}{k_s+1}\right) \int_{-\infty}^{\infty} \left(\frac{k_s-1}{p-1}\right) F_s^{p-1}(x) \{1-F_s(x)\}^{k_s-p} dF_s(x) + \\ &+ m^{-\frac{1}{2}} \sum_{j'=1}^k k_{sj'}(d_{j'}, -d_{j'}) \int_{-\infty}^{\infty} \left\{ -\Psi_s\left(\frac{1}{k_s+1}\right) [1-F_s(x)]^{k_s-2} + \Psi_s\left(\frac{k_s}{k_s+1}\right) F_s^{k_s-2}(x) + \right. \\ &\quad \left. + \sum_{p=2}^{k_s-1} \Psi_s\left(\frac{p}{k_s+1}\right) \left[\left(\frac{k_s-2}{p-2}\right) F_s^{p-2}(x) \{1-F_s(x)\}^{k_s-p} \right. \right. \\ &\quad \left. \left. - \left(\frac{k_s-2}{p-1}\right) F_s^{p-1}(x) \{1-F_s(x)\}^{k_s-p-1} \right] \right\} f_s(x) dF_s(x) + o(m^{-\frac{1}{2}}) \end{aligned}$$

by writing x for $x+d_{j'}/m^{\frac{1}{2}}$ (cf. eq. (1)), and after some calculation

$$(20) \quad = \bar{\Psi}_s + m^{-\frac{1}{2}} \cdot \sum_{j'=1}^k k_{sj'}(d_{j'}, -d_{j'}) Q_s + o(m^{-\frac{1}{2}})$$

with Q_s defined by (17). (16) now follows from (2), (3), (4) and (6).

Remark

From (2), (3), (4) and (20) it follows that

$$(21) \quad E(\tilde{t}_{sj}|H) = \sum_h [E(\Psi_{sjh}|H) - \bar{\Psi}_s] = o(m^{-\frac{1}{2}}).$$

Lemma 2: For all $j, j'=1, 2, \dots, k$

$$\begin{aligned} \text{var}(t_j|H) &= \text{var}(t_j|H_0) + o(1) \\ \text{and} \\ \text{cov}(t_j, t_{j'}|H) &= \text{cov}(t_j, t_{j'}|H_0) + o(1), \quad j \neq j' \end{aligned}$$

where the remainder terms tend to zero as $m \rightarrow \infty$.

Proof: Let, for $j \neq j'$ and/or $h \neq h'$,

$$P_{p,q}(x,y) = P(r_{sjh}=p, r_{sj'h'}=q | x_{sjh}=x, x_{sj'h'}=y, H),$$

then

$$\begin{aligned} E(\Psi_{sjh} \Psi_{sj'h'}|H) &= \sum_{p \neq q} \sum_{q=1}^{k_s} \Psi_s\left(\frac{p}{k_s+1}\right) \Psi_s\left(\frac{q}{k_s+1}\right) P(r_{sjh}=p, r_{sj'h'}=q | H) \\ &= 2 \sum_{\substack{p,q=1 \\ (p < q)}}^{k_s} \Psi_s\left(\frac{p}{k_s+1}\right) \Psi_s\left(\frac{q}{k_s+1}\right) \int_{x < y} P_{p,q}(x,y) dF_{sj}(x) dF_{sj'}(y) \end{aligned}$$

and similar as in Stoker (1955) p. 101-103 by means of Van Dantzig's method of collective marks (cf. Van Dantzig (1947-1950)) it follows for $0 < A_1, A_2 < 1$ that

$$\sum_{p,q=1}^k \sum_{p < q} P_{p,q} (x,y) A_1^{p-1} A_2^{q-1} = A_2 [A_1 A_2 F_{sj}(x) + A_2 \{F_{sj}(y) - F_{sj}(x)\} + 1 - F_{sj}(y)]^{k_{sj}-2}$$

$$\prod_{j=1}^k [A_1 A_2 F_{sj}(x) + A_2 \{F_{sj}(y) - F_{sj}(x)\} + 1 - F_{sj}(y)]^{k_{sj}}$$

assuming $j=j'$ and $h \neq h'$, and similar for the other cases,

$$= A_2 [A_1 A_2 F_s(x) + A_2 \{F_s(y) - F_s(x)\} + 1 - F_s(y)]^{k_s-2} + o(1)$$

since, from (1), $F_{sj}(x) = F_s(x) + o(1)$ where $o(1) \xrightarrow{p} 0$.

Consequently

$$E(\psi_{sjh} \psi_{sj'h'} | H) = 2 \sum_{p,q=1}^k \sum_{p < q} \psi_s \left(\frac{p}{k_s+1} \right) \psi_s \left(\frac{q}{k_s+1} \right) \int_{x < y} \frac{(k_s^*-2)!}{(p-1)!(q-p-1)!(k_s-q)!} \cdot$$

$$\cdot F_s^{p-1}(x) \{F_s(y) - F_s(x)\}^{q-p-1} \{1 - F_s(y)\}^{k_s-q} dF_s(x) dF_s(y) + o(1)$$

$$= k_s^{-1} (k_s-1)^{-1} \sum_{p \neq q} \psi_s \left(\frac{p}{k_s+1} \right) \psi_s \left(\frac{q}{k_s+1} \right) + o(1)$$

from the theory of the Dirichlet-distribution (cf. Wilks (1962) p. 177),

$$= k_s^{-1} (k_s-1)^{-1} \left[k_s^2 \psi_s^2 - \sum_p \psi_s^2 \left(\frac{p}{k_s+1} \right) \right] + o(1),$$

so that from (9) and (20) for $j \neq j'$ and/or $h \neq h'$

$$\text{cov}(\psi_{sjh}, \psi_{sj'h'} | H) = -(k_s-1)^{-1} \sigma_s^2 + o(1).$$

Further, similar to the proof of lemma 1,

$$E(\psi_{sjh}^2 | H) = k_s^{-1} \sum_p \psi_s^2 \left(\frac{p}{k_s+1} \right) + o(1),$$

so that from (9) and (20)

$$\text{var}(\psi_{sjh} | H) = \sigma_s^2 + o(1).$$

Hence, from (2), (3), (7) and (8)

$$\text{var}(\tau_{sj} | H) = \text{var}(\tau_{sj} | H_0) + o(1) \text{ and}$$

$$\text{cov}(t_{sj}, t_{sj} | H) = \text{cov}(t_{sj}, t_{sj} | H_0) + o(1), \quad j \neq j'.$$

Since t_{sj} for $s=1, 2, \dots, m$ are independent,

$$|\text{var}(t_j | H) - \text{var}(t_j | H_0)| \leq m^{-1} \sum_s |\text{var}(t_{sj} | H) - \text{var}(t_{sj} | H_0)|$$

$$\xrightarrow{m} 0$$

and similar for the covariances, which concludes the proof.

We now proceed with the proof of part (ii) of the theorem:

Assuming H to be true, we have for $0 < \delta \leq 2$

$$\begin{aligned} E|t_j|^{2+\delta} &\leq E(t_j^4) = E(m^{-1} \sum_s t_{sj}^4) \\ &= m^{-2} E(\sum_{ss'} \sum_{s''s'''} t_{sj} t_{s'j} t_{s''j} t_{s'''j}) \end{aligned}$$

where all the indices s take on the values $1, 2, \dots, m$. Since t_{sj} and $t_{s'j}$ are independent for $s \neq s'$, it immediately follows from (19) and (21) that $E(t_j^4 | H) = 0(1)$.

Hence, if the variance-covariance matrix of $t_1, \dots, t_{k-1}$ under H is denoted by M , $\vec{t}' \Lambda^{-1} \vec{t}$ and $\vec{t}' M^{-1} \vec{t}$ have the same limiting distribution under H , since

$$\begin{aligned} E[|(\vec{t}' \Lambda^{-1} \vec{t} - \vec{t}' M^{-1} \vec{t})| | H] &\leq E \sum_{\xi \xi'} \left| \frac{\Lambda_{\xi \xi'}}{|\Lambda|} - \frac{M_{\xi \xi'}}{|M|} \right| \cdot |t_\xi t_{\xi'}| \\ &\leq \sum_{\xi \xi'} \left| \frac{\Lambda_{\xi \xi'}}{|\Lambda|} - \frac{M_{\xi \xi'}}{|M|} \right| \sqrt{E t_\xi^2 \cdot E t_{\xi'}^2} \\ &\xrightarrow{m} 0 \end{aligned}$$

from lemma 2 and since $|\Lambda| > 0$ for m large.

Part (ii) of the theorem now follows in a similar way as the first part except that T in the present case has asymptotically, for $m \rightarrow \infty$,

a non-central χ^2 -distribution with non-centrality parameter

$$\lambda^2 = \lim_{m \rightarrow \infty} \vec{\mu}' M^{-1} \vec{\mu} \quad \text{where} \quad \vec{\mu}' = (\mu_1, \dots, \mu_{k-1}) \quad \text{with} \quad \mu_\xi = E(t_\xi | H) \quad \text{given by}$$

$$= \sum_{\xi} \sum_{\xi'} \lim_{m \rightarrow \infty} \frac{|M_{\xi\xi'}|}{|M|} E(t_{\xi}|H) E(t_{\xi'}|H),$$

whence (15) follows from lemma 2.

Remarks

1. It is implicitly assumed that the limit, for $m \rightarrow \infty$, in equation (15) exists. This is true in the special case considered in par. 5 and in part II.
2. It may be pointed out that if $F_s(x) = F(x + e_s)$, with $\{e_s\}$ a set of finite real constants, then

$$(22) \quad \int_{-\infty}^{\infty} F_s^{r-1}(x) f_s^2(x) dx = \int_{-\infty}^{\infty} F^{r-1}(x) f^2(x) dx, \text{ that is, additive}$$

block effects have no effect on the non-centrality parameter λ^2 .

3. Condition (13) can be replaced by the following somewhat weaker

condition, viz. the matrix $\{\kappa_{jj'}\}$, with $\kappa_{jj'} = \lim_{m \rightarrow \infty} m^{-1} \sum_s \kappa_{sj} \kappa_{sj'}$,

must not be a matrix of the type $\begin{pmatrix} P & 0' \\ 0 & Q \end{pmatrix}$ in which P and Q are square matrices and 0 and $0'$ consist of zeros only provided

$\lim_{m \rightarrow \infty} m^{-1} \sum_s \kappa_{sj}$ exists and is > 0 for all j (cf. Benard and Van Elteren (1953) p. 363).

5. The R B D with c observations per cell

5.1 The statistic T_c

In the special case where all $\kappa_{sj} = c$ so that $\kappa_s = kc$ for all s , the test statistic T is written as T_c . Then $\Psi_s(\delta)$ and σ_s^2 become independent of s and can be written as $\Psi(\delta)$ and σ^2 respectively.

Let

$$(23) \quad A = c^2 \sigma^2 / (kc - 1),$$

then it follows from (10) and (11) that

$$(24) \quad \text{var}(t_j | H_0) = (k-1)A$$

and

$$(25) \quad \text{cov}(t_j, t_{j'} | H_0) = -A, \quad j \neq j',$$

both independent of m .

It readily follows from (24) and (25), with Λ the variance-covariance matrix of $t_1, \dots, t_{k-1}$ (cf. par. 2), that

$$(26) \quad |\Lambda| = A^{k-1} k^{k-2} > 0$$

and

$$|\Lambda_{\xi\xi'}| = \begin{cases} 2A^{k-2}k^{k-3} & \text{for } \xi=\xi' \\ A^{k-2}k^{k-3} & \text{for } \xi \neq \xi'. \end{cases}$$

Consequently,

$$(27) \frac{|\Lambda_{\xi\xi'}|}{|\Lambda|} = \begin{cases} \frac{2}{Ak} & \text{for } \xi=\xi' \\ \frac{1}{Ak} & \text{for } \xi \neq \xi', \end{cases}$$

independent of m , so that

$$\begin{aligned} (28) \quad T_c &= \sum_{\xi=1}^{k-1} \sum_{\xi'=1}^{k-1} \frac{|\Lambda_{\xi\xi'}|}{|\Lambda|} t_{\xi} t_{\xi'} \\ &= \frac{2}{Ak} \sum_{\xi} t_{\xi}^2 + \frac{1}{Ak} \sum_{\xi \neq \xi'} t_{\xi} t_{\xi'} \\ &= \frac{1}{Ak} \sum_{j=1}^k t_j^2 \quad \text{since } \sum_{j=1}^k t_j = 0 \\ &= \frac{kc-1}{kc^2\sigma^2} \sum_{j=1}^k t_j^2 \end{aligned}$$

$$\text{where } t_j = m^{-\frac{1}{2}} \sum_{s=1}^m \sum_{h=1}^c \{ \psi(\frac{r_{sjh}}{kc+1}) - \bar{\psi} \}.$$

Theorem 2

T_c , defined by eq. (28), has asymptotically, for $m \rightarrow \infty$,

(i) a (central) χ^2 -distribution under H_0 and

(ii) a non-central χ^2 -distribution under H ,

with $k-1$ degrees of freedom, provided the additional condition (14) holds in case (ii). The non-centrality parameter of the limiting non-central χ^2 -distribution is given by

$$(29) \quad \lambda_c^2 = \frac{kc-1}{kc^2\sigma^2} \lim_{m \rightarrow \infty} \left\{ 2 \sum_{\xi} [E(t_{\xi}|H)]^2 + \sum_{\xi \neq \xi'} E(t_{\xi}|H) E(t_{\xi'}|H) \right\}$$

where

$$(30) \quad E(t_j|H) = c^2 \sum_{j'=1}^k (d_{j'} - d_j) R_m + o(1)$$

with

$$(31) \quad R_m = \sum_{p=1}^{kc} \psi\left(\frac{p}{kc+1}\right) \sum_{\substack{r=p-1 \\ (\neq 0)}}^{kc-1} (-1)^{r-p+1} \binom{kc-2}{r-1} \binom{r}{p-1} \frac{1}{m} \sum_s \int_{-\infty}^{\infty} F_s^{r-1}(x) f_s^2(x) dx.$$

Proof

Since $k_s = k \cdot c < \infty$ and $m^{-1} \sum_s k_{sj} k_{sj'} = c^2 > 0$ for all j, j' , conditions

(12) and (13) of theorem 1 hold and theorem 2 immediately follows from theorem 1 by making use of (23) and (27).

Remarks

1. If the set of values $\{d_j\}$ satisfies the relation

$$(32) \sum_j d_j = 0,$$

then we obtain from (30)

$$E(t_j | H) = -d_j k c^2 R_m + o(1)$$

so that

$$\sum_{j=1}^k E(t_j | H) = o(1).$$

Consequently

$$(33) \lambda_c^2 = \frac{kc-1}{kc^2\sigma^2} \sum_{j=1}^k \lim_{m \rightarrow \infty} [E(t_j | H)]^2$$

$$= \frac{k(kc-1)c^2}{\sigma^2} \sum_{j=1}^k d_j^2 \lim_{m \rightarrow \infty} R_m^2.$$

2. In the case of additive block effects, i.e. $F_s(x) = F(x + e_s)$ with $\{e_s\}$ a set of finite real constants, (29) or (33) becomes independent of m and depends only on Ψ, k, c and F . (cf. remark 2 at the end of par. 4).

5.2. Asymptotic Relative Efficiencies

5.2.1. Introduction

The a.r.e. of the test based on T_c with respect to the F-test is given by

$$(34) E_{T_c, F} = \lambda_c^2 / \lambda_F^2$$

where λ_F^2 denotes the non-centrality parameter of the F-test. In the case of additive block effects, i.e.

$$(35) F_{sj}(x) = F(x + e_s + d_j / \sqrt{m}) \text{ for all } x, \text{ with } \sum_j d_j = 0,$$

it can be shown that (cf. Tang (1938))

$$(36) \lambda_F^2 = c \sigma_F^{-2} \sum_j d_j^2$$

where σ_F^2 is the variance associated with $F(x)$.

Note that under the class of alternatives (35) it follows from (33), remark 2 of par. 5.1 and from (34) and (36) that $E_{T_c, F}$ does not depend on k and c separately, but only on the product kc .

5.2.2. $\Psi(\delta) \equiv \delta$.

We now have $\bar{\Psi} = \frac{1}{2}$ and $\sigma^2 = \frac{kc-1}{12(kc+1)}$, so that

$$T_c = \frac{12}{mkc^2(kc+1)} \sum_{j=1}^k \left[\sum_{s=1}^m \sum_{h=1}^c \{r_{sjh} - \frac{1}{2}(kc+1)\} \right]^2.$$

Although in this special case the value of λ_c^2 can easily be obtained from first principles (e.g. analogously to Van Elteren and Noether (1959)), we nevertheless derive it from the above formulae.

From (31) we have

$$\begin{aligned} R_m &= \left[\sum_{p=1}^2 \sum_{r=p-1}^1 + \sum_{r=2}^{kc-1} \sum_{p=1}^{r+1} \right] \left(\frac{p}{kc+1} \right) (-1)^{r-p+1} \binom{r}{p-1} \binom{kc-2}{r-1} \\ &\quad \cdot \frac{1}{m} \sum_s f_{\infty}^{r-1}(x) f_s^2(x) dx \\ &= \sum_{p=1}^2 \left(\frac{p}{kc+1} \right) (-1)^{2-p} \binom{1}{p-1} \frac{1}{m} \sum_s \int_{-\infty}^{\infty} f_s^2(x) dx \quad \text{from the appendix} \\ &= (kc+1)^{-1} m^{-1} \sum_s \int_{-\infty}^{\infty} f_s^2(x) dx. \end{aligned}$$

Consequently, from (33) (assuming $\sum_j d_j = 0$),

$$\lambda_c^2 = \frac{12kc^2}{kc+1} \sum_{j=1}^k d_j^2 \lim_{m \rightarrow \infty} \left[\frac{1}{m} \sum_s \int_{-\infty}^{\infty} f_s^2(x) dx \right]^2,$$

which, in the case of additive block effects, reduces to (cf. remark 2 of par. 5.1)

$$\lambda_c^2 = \frac{12kc}{kc+1} \sum_{j=1}^k d_j^2 \left[\int_{-\infty}^{\infty} f^2(x) dx \right]^2.$$

Hence the a.r.e. of the test based on T_c with respect to the F-test under the class of alternatives (35) is from (34) and (36)

$$(37) E_{T_c, F} = 12kc(kc+1)^{-1} \sigma_F^2 \left[\int_{-\infty}^{\infty} f^2(x) dx \right]^2.$$

This is the same result as that obtained by Van Elteren and Noether (1959) for the case $c=1$.

If F is the c.d.f. of the normal distribution, (37) becomes

$$(38) E_{T_c, F} = \frac{3}{\pi} \cdot \frac{kc}{kc+1}$$

which approaches the value $\frac{3}{\pi} = 0.955$ when kc becomes large,

and if F is the c.d.f. of the homogeneous distribution, we have

$$E_{T_c, F} = \frac{kc}{kc+1},$$

which approaches the value 1 when kc becomes large.

Remark: In the same way the value of λ_c^2 can be obtained when $\Psi(\delta)$ is a polynomial in δ . For instance, if $\Psi(\delta) = \delta^2$, we obtain (assuming $\sum_j d_j = 0$)

$$\lambda_c^2 = \frac{180kc^2}{(2kc+1)(8kc+11)(kc+1)} \sum_j d_j^2 \lim_{m \rightarrow \infty} \left[\frac{1}{m} \sum_s \{ 3 \int_{-\infty}^{\infty} f_s^2(x) dx + 2(kc-2) \right. \\ \left. \cdot \int_{-\infty}^{\infty} F_s(x) f_s^2(x) dx \} \right]^2.$$

$$5.2.3. \quad \Psi(\delta) \equiv Q(\delta) \equiv \Phi^{-1}(\delta)$$

where $\Phi(x)$ is the c.d.f. of the standard normal distribution - cf. Van der Waerden (1957).

In this case we have

$$\bar{\Psi} = 0 \text{ and } \sigma^2 = (kc)^{-1} \sum_{p=1}^{kc} Q^2\left(\frac{p}{kc+1}\right),$$

a function which is tabled in Van der Waerden (1957), so that

$$T_c = \frac{kc-1}{cm \sum_p Q^2\left(\frac{p}{kc+1}\right)} \sum_{j=1}^k \left[\sum_{s=1}^m \sum_{h=1}^c Q\left(\frac{r_{sjh}}{kc+1}\right) \right]^2.$$

From (31), (33), remark 2 of par. 5.1, (34) and (36) we obtain for the class of alternatives (35)

$$(39) E_{T_c, F} = \frac{k^2 c^2 (kc-1) \sigma_F^2 R^2}{kc \sum_{p=1}^{kc} Q^2\left(\frac{p}{kc+1}\right)}$$

with

$$(40) R = \sum_{p=1}^{kc} Q\left(\frac{p}{kc+1}\right) \sum_{\substack{r=p-1 \\ (\neq 0)}}^{kc-1} (-1)^{r-p+1} \binom{kc-2}{r-1} \binom{r}{p-1} \int_{-\infty}^{\infty} F^{r-1}(x) f^2(x) dx.$$

No general simplification of R could be obtained.

Normal distribution

If F is the c.d.f. of the standard normal distribution, we have $\sigma_F^2 = 1$. $E_{T_C, F}$ has only been calculated for the following values of kc :-

kc	$E_{T_C, F}$	kc	$E_{T_C, F}$
2	0.6366	8	0.8566
3	0.7162	9	0.8687
4	0.7651	10	0.8788
5	0.7985	11	0.8873
6	0.8230	12	0.8946
7	0.8417	13	0.9010

Remark: If these values are compared with the values obtained from (38) for the different values of kc , we observe that the a.r.e. values given by (39) are slightly higher (except when $kc=2$) than those given by (38) and, moreover, the ratio of the a.r.e. value given by (38) to that given by (39) decreases from 1 for $kc=2$ to 0.9841 for $kc=13$.

Note that for $kc=2$ (i.e. $k=2, c=1$) the test based on T_C is in fact the two-sided sign test irrespective of the special function $\Psi(\delta)$ chosen. (Cf. Van Elteren and Noether (1959) for a discussion on the a.r.e. of the one and two-sided sign tests.)

If further $kc \rightarrow \infty$ (with k fixed and $c \rightarrow \infty$) it seems obvious to expect that $E_{T_C, F}$ given by (39) should tend to 1 by making use of results

obtained by Chernoff and Savage (1958) for the two-sample problem and which have been extended to the problem of k samples by Lemmer (1964) and Puri (1964). However, we have not succeeded in proving that $\lim_{kc \rightarrow \infty} (kcR)^2 = \sigma_F^2$, which should then be the case.

Homogeneous distribution

If F is the c.d.f. of the homogeneous distribution, $\sigma_F^2 = \frac{1}{12}$ and $\int_{-\infty}^{\infty} F^{p-1}(x) f^2(x) dx = \frac{1}{p}$. We thus have from (40)

$$(41) R = (kc-1)^{-1} \sum_{p=1}^{kc} Q\left(\frac{p}{kc+1}\right) \binom{kc-1}{p-1} \sum_{r=p-1}^{kc-1} (-1)^{r-p+1} \binom{kc-p}{r-p+1} \quad (\neq 0)$$

$$= (kc-1)^{-1} \left[-Q\left(\frac{1}{kc+1}\right) + Q\left(\frac{kc}{kc+1}\right) \right]$$

$$= 2(kc-1)^{-1} Q\left(\frac{kc}{kc+1}\right) \quad \text{since } Q \text{ is symmetric about zero.}$$

From (39) follows

$$(42) E_{T_c, F} = \frac{kc}{3(kc-1)} \cdot \frac{1}{\frac{1}{kc} \sum_{p=1}^{kc} Q^2\left(\frac{p}{kc+1}\right)} \cdot Q^2\left(\frac{kc}{kc+1}\right).$$

Remarks:

1. In the case of the homogeneous distribution, the value of R given by (41) holds true for any arbitrary function $\Psi(\delta)$ (replacing $Q(\delta)$ by $\Psi(\delta)$).
2. For kc large, Stoker (1955) proved that (cf. p. 57)

$$Q\left(\frac{kc}{kc+1}\right) = \sqrt{2 \ln kc} + o((\ln kc)^{-\frac{1}{2}+\epsilon}), \quad 0 < \epsilon < \frac{1}{2}$$

and

$$\lim_{kc \rightarrow \infty} \frac{1}{kc} \sum_p Q^2\left(\frac{p}{kc+1}\right) = 1,$$

so that $E_{T_c, F}$ can be made arbitrary large by increasing kc .

P A R T II

6. The balanced incomplete block design (BIBD)

6.1 Notation

We now consider a two-way layout in a BIBD and suppose that the k treatment combinations consist of two components A and B at $a(\geq 2)$ and $b(\geq 2)$ levels respectively, so that $k=ab$.

Let

- t = number of treatment combinations (plots) occurring in each block ($t \leq k$). No treatment combination is replicated in the same block;
- n = number of replications of each treatment combination;
- m = number of blocks = kn/t ;
- λ = number of times that any pair of treatment combinations occurs in the same block = $n(t-1)/(k-1)$.

The class of hypotheses (1) is now reformulated as

$$(43) H : F_{sij}(x) = F_s(x + d_{ij}/n^{\frac{1}{2}}) \text{ for all real } x^*$$

*) Throughout part II it is assumed that the indices $i, i'=1, 2, \dots, a$; $j, j'=1, 2, \dots, b$; $h, h'=1, 2, \dots, n$; $v, v'=1, 2, \dots, t$ and $r, r'=1, 2, \dots, k$, whereas the indices s and ξ assume the same set of values as in part I.

where $F_{sij}(x)$ denotes the c.d.f. of the population from which the observation of the (i,j) -th treatment combination in the s -th block is drawn assuming that the (i,j) -th treatment combination is applied to a plot in the s -th block, all $F_s(x)$ are assumed continuous and

$$(44) \begin{cases} d_{ij} = a_i + b_j + c_{ij}, \text{ the } a_i \text{'s, } b_j \text{'s and } c_{ij} \text{'s being arbitrary real} \\ \text{numbers satisfying the relations} \\ \sum_i a_i = 0 = \sum_j b_j, \sum_i c_{ij} = 0 = \sum_j c_{ij}. \end{cases}$$

Also

$$H_0 : F_{sij}(x) = F_s(x) \text{ for all real } x \text{ and all } s, i \text{ and } j.$$

Let x_{ijh} denote the result of the h -th replication of the (i,j) -th treatment combination and r_{ijh} the rank allocated to x_{ijh} in the block containing x_{ijh} . (For the adjustments in notation necessary in the case when ties among the observations are admissible - cf. Reinach (1965) and Lemmer and Stoker (1967)).

If now $\delta_{ijh} = r_{ijh}/(t+1)$, $1 \leq r_{ijh} \leq t$ for all i, j and h , and $z_{ijh} = \Psi(\delta_{ijh})$, define.

$$z_{ij.} = n^{-1} \sum_h z_{ijh}, z_{i..} = b^{-1} \sum_j z_{ij.}, z_{.j.} = a^{-1} \sum_i z_{ij.},$$

$$z_{...} = (abn)^{-1} \sum_{ijh} z_{ijh} = t^{-1} \sum_{ijh} \Psi \left[\frac{v}{(t+1)} \right] = \bar{\Psi}, \text{ say,}$$

and

$$\sigma^2 = (abn)^{-1} \sum_{ijh} (z_{ijh} - z_{...})^2 = t^{-1} \sum_{ijh} \left\{ \Psi \left(\frac{v}{t+1} \right) - \bar{\Psi} \right\}^2$$

From our assumptions it follows that $\sigma^2 > 0$, independent of n or m .

Let further

$$(45) \begin{aligned} t_{ij} &= C n^{\frac{1}{2}} (z_{ij.} - z_{...}) \\ u_{ij} &= C n^{\frac{1}{2}} (z_{ij.} - z_{i..} - z_{.j.} + z_{...}) \\ u_{i.} &= C n^{\frac{1}{2}} b^{\frac{1}{2}} (z_{i..} - z_{...}) \\ u_{.j} &= C n^{\frac{1}{2}} a^{\frac{1}{2}} (z_{.j.} - z_{...}) \end{aligned}$$

where

$$(46) C = (k-1)^{\frac{1}{2}} k^{-\frac{1}{2}} \sigma^{-1}.$$

The following test statistics are now defined:

$$\text{For total S.S.:} \quad T_T = \sum_{ij} t_{ij}^2$$

For S.S. for component A: $T_A = \sum_i u_i^2$
(47)

For S.S. for component B: $T_B = \sum_j u_{.j}^2$

For S.S. for interaction: $T_{AB} = \sum_{ij} u_{ij}^2$
 $= T_T - T_A - T_B.$

6.2 Limiting distribution of the test statistics

Theorem 3

T_A , T_B and T_{AB} , defined by (47), are asymptotically, for $n \rightarrow \infty$,^{*)} distributed independently

(i) as (central) χ^2 -variates under H_0 and

(ii) as non-central χ^2 -variates under H (defined by (43) and (44)), with $(a-1)$, $(b-1)$ and $(a-1)(b-1)$ degrees of freedom respectively, provided the following additional condition holds in case (ii), viz.

(48) all $F_g(x) \equiv F(x + e_g)$, with $\{e_g\}$ a set of finite real constants, where $f(x)$ has a density function $f(x)$ which is assumed continuous and uniformly bounded in x .

The non-centrality parameters of the non-central χ^2 -distributions are given by

$$(49) \quad \lambda_A^2 = C^2 b M^2 \sum_i a_i^2, \quad \lambda_B^2 = C^2 a M^2 \sum_j b_j^2 \quad \text{and} \quad \lambda_{AB}^2 = C^2 M^2 \sum_{ij} c_{ij}^2$$

respectively, where

$$(50) \quad M = \frac{k(t-1)}{k-1} \sum_{v=1}^t \Psi\left(\frac{v}{t+1}\right) \sum_{\ell=v-1}^{t-1} (-1)^{\ell-v+1} \binom{\ell}{v-1} \binom{t-2}{\ell-1} \int_{-\infty}^{\infty} F^{\ell-1}(x) f^2(x) dx. \\ (\neq 0)$$

Proof^{xx)}

In a similar way as in par. 5.1 of part I it can easily be shown that T_T is a special case of the general statistic T defined by eq.

(5) (cf. Reinach (1966)). Note that the number of observations k_{sij} of the (i,j) -th treatment combination occurring in the s -th block is in a BIBD either 1 or 0 depending on whether this treatment combination is applied to a plot in the s -th block or not. Further, for all s (i.e. blocks), $\sum_i \sum_j k_{sij} = t(\leq k)$.

Since, for $i \neq i'$ and/or $j \neq j'$,

^{*)} It is assumed that k and t remain finite.

^{xx)} An alternative proof of case (i), without making use of Cochran's theorem, is given by Reinach (1966).

$$\lim_{m \rightarrow \infty} m^{-1} \sum_s k_{sij} k_{si'j'} = \lambda/m = \frac{t(t-1)}{k(k-1)} > 0 \quad (\text{for } t \geq 2),$$

it follows that the conditions of theorem 1 are satisfied, so that (cf. also theorem 2) T_T has asymptotically a central χ^2 -distribution under H_0 and a non-central χ^2 -distribution under the class of hypotheses defined by (43), (44) and (48), with $(k-1)$ degrees of freedom.

In the non-central case the non-centrality parameter is given by

$$(51) \lambda_T^2 = C^2 M^2 \sum_{ij} d_{ij}^2$$

with C defined by (46) and M by (50), since from (16), (17) and (22) we now have

$$E(t_{ij}|H) = C n^{-1} \sum_{s=1}^m k_{sij} \sum_{i'=1}^a \sum_{j'=1}^b k_{si'j'} (d_{i'j'} - d_{ij}) Q + o(1)$$

with $\frac{k(t-1)}{k-1} Q = M$.

Consequently, since $\sum_{ij} d_{ij} = 0$,

$$\begin{aligned} (52) \quad E(t_{ij}|H) &= C n^{-1} \lambda \sum_{i'j'} (d_{i'j'} - d_{ij}) Q + o(1) \\ &= -C n^{-1} k \lambda d_{ij} Q + o(1) \\ &= -C M d_{ij} + o(1). \end{aligned}$$

Furthermore, $\sum_{ij} E(t_{ij}|H) = o(1)$ from (44),

so that similar to the derivation given in par. 5.1 (cf. also remark 1),

$$\lambda_T^2 = \lim_{n \rightarrow \infty} \sum_{ij} [E(t_{ij}|H)]^2,$$

from which (51) follows.

The proof of the theorem can now be completed in exactly the same way as in Lemmer and Stoker (1967) by making use of Cochran's theorem for the non-central case (cf. Graybill (1961) theorem 4.16). In order to obtain the non-centrality parameters of the limiting non-central χ^2 -distributions, we observe that, since $\sum_i u_{i.} = 0$, T_A can be written

(cf. Lemmer and Stoker (1967))

$$T_A = \{\vec{u}_{i.}\}' \{\delta_{ii}, -a^{-1}\} \{\vec{u}_{i.}\} \quad \text{with } \{\vec{u}_{i.}\}' \text{ a row vector consisting of}$$

the elements $u_{i.}$ and with δ_{ii} , denoting the Kronecker delta. Thus from Cochran's theorem

$$\begin{aligned}
\lambda_A^2 &= \lim_{n \rightarrow \infty} \{E_H \vec{u}_{i.}\}' \{\delta_{ii}, -a^{-1}\} \{E_H \vec{u}_{i.}\} \text{ with } E_H \equiv E(\cdot | H) \\
&= \lim_{n \rightarrow \infty} \sum_i (E_H u_{i.})^2 \\
&= b C^2 M^2 \Sigma a_i^2, \text{ as given by (49),}
\end{aligned}$$

since

$$\begin{aligned}
E_H u_{i.} &= b^{\frac{1}{2}} E_H (b^{-1} \Sigma t_{ij}) \text{ from (45) and by making use of (44) and} \\
(52). \\
&\text{Similarly for } \lambda_B^2 \text{ and } \lambda_{AB}^2.
\end{aligned}$$

Remark: From (49) and (51) we observe that

$$\lambda_T^2 = \lambda_A^2 + \lambda_B^2 + \lambda_{AB}^2,$$

as it should be.

6.3 The single degree of freedom approach

Define $k-1$ sets of orthogonal constants $\{g_{r,\xi}\}$, $r=1, \dots, k$; $\xi=1, \dots, k-1$, satisfying

$$\begin{aligned}
&\sum_r g_{r,\xi} = 0 \\
(53) \quad &\text{and} \\
&\sum_r g_{r,\xi} g_{r,\xi'} = 0, \quad \xi \neq \xi'.
\end{aligned}$$

Let for all r

$$\begin{aligned}
e_{r,\xi} &= g_{r,\xi} / (\sum_r g_{r,\xi}^2)^{\frac{1}{2}}, \quad \xi=1, 2, \dots, k-1 \\
(54) \quad e_{r,k} &= k^{-\frac{1}{2}}
\end{aligned}$$

and define $E = \{e_{r,\xi}\}$. Then E is an orthogonal $k \times k$ matrix (cf. Reinach (1966) p. 49).

$$\begin{aligned}
&\text{If} \\
(55) \quad L_\xi &= \sum_{r=1}^k e_{r,\xi} t_r, \quad \xi=1, 2, \dots, k-1,
\end{aligned}$$

with

$$t_r = t_{ij}, \quad r=(i-1)b+j \text{ for all } i \text{ and } j,$$

then

$$\sum_{\xi=1}^{k-1} L_\xi^2 = \{\vec{t}_r\}' \{\delta_{rr}, -k^{-1}\} \{\vec{t}_r\} = T_T$$

since

$$(56) \sum_{\xi=1}^{k-1} e_{r,\xi} e_{r',\xi} = \delta_{rr'} k^{-1}$$

due to the orthogonality of E .

But

$$(57) T_{\xi} \equiv L_{\xi}^2 = \{\vec{t}_r\}' \{e_{r,\xi} e_{r',\xi}\} \{\vec{t}_r\}, \text{ a non-negative quadratic form of rank 1 in the } t_r \text{'s.}$$

Similar to the proof of theorem 3 the following now immediately follows from Cochran's theorem for the non-central case, viz.

Theorem 4

$T_1, \dots, T_{k-1}$ defined by (57) are asymptotically, for $n \rightarrow \infty$, distributed independently.

(i) as (central) χ^2 -variates under H_0 and

(ii) as non-central χ^2 -variates under H ,

with one degree of freedom each, provided the additional condition (48) holds in case (ii). The non-centrality parameter of the limiting distribution of T_{ξ} under H is given by

$$\lambda_{\xi}^2 = C^2 M^2 \sum_{r,r'} e_{r,\xi} e_{r',\xi} d_r d_{r'},$$

where

$$d_r = d_{ij}, \quad r = (i-1)b+j.$$

Remark: Note that

$$\begin{aligned} \sum_{\xi=1}^{k-1} \lambda_{\xi}^2 &= C^2 M^2 \sum_r d_r^2 \text{ from (56), since } \sum_r d_r = \sum_{ij} d_{ij} = 0, \\ &= \lambda_T^2. \end{aligned}$$

6.4 A.r.e. with respect to the parametric F-test

The non-centrality parameter $\lambda_{F_T}^2$ of the limiting non-central χ^2 -distribution (for the number of replications tending to infinity) of the test statistic F_T of the parametric F-test for the total S.S. in the case of the BIBD under the class of hypotheses defined by (43), (44) and (48), is given by

$$\lambda_{F_T}^2 = \frac{k(t-1)}{t(k-1)} \frac{\sum_{ij} d_{ij}}{\sigma_F^2}$$

where σ_F^2 denotes the variance associated with $F(x)$ (cf. Van Elteren and Noether (1959) eq. (3)).

By making use of this result it can easily be shown that the a.r.e. of the test based on T_T with respect to the parametric F-test for the total S.S. reduces to exactly the same expression as

that given by equation (34), in the case of additive block effects for the test based on the test-statistic T_c with only kc replaced by t . Consequently, all results obtained in par. 5.2 for the a.r.e. in different special cases apply here.

The same applies to the tests based on the test statistics T_A , T_B , T_{AB} and T_ξ with respect to the corresponding parametric F-tests.

6.5 Concluding Remarks

- (a) The theory derived in par. 6.2 and 6.3 still holds true in the non-central case if condition (48) is replaced by the more general condition (14). The expressions for the non-centrality parameters, however, become more complicated since they, and therefore the power of the tests, will depend on which particular treatment combinations are applied to plots in which particular block (cf. the proof of theorem 3 and also Van Elteren and Noether (1959)). This is not the case for the randomised block design considered in part I, par. 5.
- (b) The applicability of the tests: The large sample χ^2 -approximation to the exact distributions of the test statistics and the applicability of the tests under alternative hypotheses are discussed in Reinach (1965) and (1966) for the randomised block design ($k=t$, $n=m=\lambda$) in the special case where $\Psi(\delta) \equiv \delta$ (in this case T_T reduces to Friedman's (1937) test statistic for the one-way classification).

Undesirable consistency properties exist, however, in the tests for the main effects if interaction effects exist (cf. Reinach (1966) p. 264 and the remark given in Reinach's appendix to Lemmer and Stoker (1967)).

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A P P E N D I X

$$\sum_{p=1}^{r+1} (-1)^{p-1} p^s \binom{r}{p-1} = 0 \text{ for } r \geq s+1, s \geq 1, r \text{ and } s \text{ integers.}$$

Proof:

$$x^n = \sum_{r=1}^n s! r S_n^{(r)} \text{ where } S_n^{(r)}, r = 1, 2, \dots, n, \text{ denote the Stirling}$$

numbers of the second type.

$$\begin{aligned} \sum_{p=1}^{r+1} (-1)^{p-1} p^s \binom{r}{p-1} &= \sum_{t=0}^r (-1)^t (t+1)^s \binom{r}{t} \\ &= \sum_{t=0}^r (-1)^t \binom{r}{t} \sum_{u=1}^s (t+1)^! u S_s^{(u)}, s \geq 1 \end{aligned}$$

$$\begin{aligned}
&= \sum_{u=1}^s \sum_{t=0}^r (-1)^t S_s^{(u)} \frac{r!t}{t!} (t+1)^! u \\
&= \sum_{u=1}^s \sum_{t=u-1}^r (-1)^t S_s^{(u)} \frac{r!t}{(t-u+1)!} (t+1) \\
&= \sum_{u=1}^s \sum_{v=0}^{r-u+1} (-1)^{v+u-1} S_s^{(u)} \frac{r! (u+v-1)}{v!} (u+v) \\
&= \sum_{u=1}^s (-1)^{u-1} S_s^{(u)} \cdot u \cdot r! (u-1)^{r-u+1} \sum_{v \neq 0} (-1)^v \frac{(r-u+1)! v}{v!} \\
&\quad + \sum_{u=1}^s (-1)^{u-1} S_s^{(u)} r! (u-1)^{r-u+1} \sum_{v \neq 1} (-1)^v \frac{(r-u+1)! v}{(v-1)!} \\
&= \sum_{u \neq 1}^s (-1)^u S_s^{(u)} r! u \sum_{t=0}^{r-u} (-1)^t \frac{(r-u)! t}{t!} \\
&= 0 \quad \text{provided } r \geq 1 + u, \text{ or } r \geq 1 + s \text{ since } u \leq s.
\end{aligned}$$

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